SUBCHAPTER 3.4 · ROTATIONAL DYNAMICS & TURBOMACHINERY

Conservation of Angular Momentum & Turbomachinery Torque

Moment balance about strategic reduction centers $\vec{x}_0$, elimination of unknown reactions, and Euler's fundamental turbomachinery equation.

1. The Master Angular Momentum Equation

In continuum mechanics, the moment of momentum (angular momentum) of a closed material fluid system $V_f(t)$ about an arbitrary fixed point in space $\vec{x}_0$ changes at a rate equal to the resultant external moment acting on the fluid:

$$\frac{d}{dt}\left[\int_{V_f(t)} \rho[(\vec{x}-\vec{x}_0)\times\vec{v}]\,dV\right] = \sum \vec{M}_{\vec{x}_0,\text{ext}}$$

[Eq. 3.37] Angular momentum principle for a material fluid volume.

Substituting intensive angular momentum $\vec{b} = (\vec{x}-\vec{x}_0)\times\vec{v}$ into the universal Reynolds Transport Theorem yields the **Master Integral Angular Momentum Equation** for any moving control volume $V_c(t)$:

$$\mathbf{\frac{d}{dt}}\left[\int_{\mathbf{V_c(t)}} \rho[(\vec{\mathbf{x}}-\vec{\mathbf{x}}_\mathbf{0})\times\vec{\mathbf{v}}]\,\mathbf{dV}\right] + \int_{\mathbf{\Sigma_c(t)}} \rho[(\vec{\mathbf{x}}-\vec{\mathbf{x}}_\mathbf{0})\times\vec{\mathbf{v}}][(\vec{\mathbf{v}}-\vec{\mathbf{v}}_\mathbf{c})\cdot\vec{\mathbf{n}}]\,\mathbf{d\sigma} = \sum \vec{\mathbf{M}}_{\vec{\mathbf{x}}_\mathbf{0},\text{ext}}$$

[Eq. 3.38] Universal Integral Angular Momentum Conservation Equation.

The external moments $\sum \vec{M}_{\vec{x}_0,\text{ext}}$ consist of pressure forces, viscous shear forces, body forces (e.g., gravity), and mechanical holding torques applied by shaft couplings or mounting flanges:

$$\sum \vec{M}_{\vec{x}_0,\text{ext}} = -\int_{\Sigma_c} (\vec{x}-\vec{x}_0)\times(p\vec{n})\,d\sigma + \int_{\Sigma_c} (\vec{x}-\vec{x}_0)\times(\bar{\bar{\tau}}'\cdot\vec{n})\,d\sigma + \int_{V_c} \rho(\vec{x}-\vec{x}_0)\times\vec{f}_m\,dV + \vec{\mathcal{M}}_{\text{shaft}}$$

Angular Momentum: Complete Derivation Chain (Notes.pdf, Eqs. 3.37-3.38)

Step 1: Angular momentum of a fluid volume (Eq. 3.37)

Let $\vec{r} = \vec{x} - \vec{x}_0$, with $\vec{x}_0$ a fixed point of an inertial frame. A fluid particle of mass $\rho\, dV$ has angular momentum $\vec{r} \wedge \rho\vec{v}\, dV$ about $\vec{x}_0$ (Notes, p. 33), so the angular momentum of the fluid volume is $\int_{V_f} \rho\, \vec{r} \wedge \vec{v}\, dV$. Newton's second law for angular momentum states that its rate of change equals the moment of the external forces about $\vec{x}_0$. The moment of the surface force $\bar{\bar{\tau}} \cdot \vec{n}\, d\sigma$ on each element of the boundary and the moment of the body force $\rho \vec{f}_m\, dV$ on each particle give Eq. 3.37:

$$\frac{d}{dt} \int_{V_f(t)} \rho\, \vec{r} \wedge \vec{v}\, dV = \int_{\Sigma_f(t)} \vec{r} \wedge (\bar{\bar{\tau}} \cdot \vec{n})\, d\sigma + \int_{V_f(t)} \rho\, \vec{r} \wedge \vec{f}_m\, dV$$

Eq. 3.37

Step 2: Consistency with the local momentum equation (extension)

Extension, not in the Notes. For a fluid particle the mass $\rho\, dV$ is constant and $D\vec{r}/Dt = \vec{v}$ because $\vec{x}_0$ is fixed. The product rule gives

$$\frac{D}{Dt}\left[\vec{r} \wedge \rho\vec{v}\, dV\right] = \left[\vec{v} \wedge \vec{v} + \vec{r} \wedge \frac{D\vec{v}}{Dt}\right]\rho\, dV = \vec{r} \wedge \rho \frac{D\vec{v}}{Dt}\, dV$$

since $\vec{v} \wedge \vec{v} = \vec{0}$. Insert Cauchy's equation (Eq. 3.25), $\rho D\vec{v}/Dt = \nabla \cdot \bar{\bar{\tau}} + \rho\vec{f}_m$, and integrate over the volume:

$$\frac{d}{dt} \int_{V_f} \rho\, \vec{r} \wedge \vec{v}\, dV = \int_{V_f} \vec{r} \wedge (\nabla \cdot \bar{\bar{\tau}})\, dV + \int_{V_f} \rho\, \vec{r} \wedge \vec{f}_m\, dV$$

It remains to show that the first integral on the right equals the surface integral of Eq. 3.37. With the permutation symbol ϵ_{kij}, component k of the integrand is $\epsilon_{kij} r_i\, \partial\tau_{lj}/\partial x_l$. The product rule and $\partial r_i/\partial x_l = \delta_{il}$ give

$$\epsilon_{kij}\, r_i \frac{\partial \tau_{lj}}{\partial x_l} = \frac{\partial}{\partial x_l}\left(\epsilon_{kij} r_i \tau_{lj}\right) - \epsilon_{kij}\, \tau_{ij}$$

The last term vanishes because ϵ_{kij} is antisymmetric in (i, j) and τ_{ij} is symmetric (Eq. 3.19). Gauss' theorem on the first term, together with $\tau_{lj} n_l = \tau_{jl} n_l = (\bar{\bar{\tau}} \cdot \vec{n})_j$, gives

$$\int_V \vec{r} \wedge (\nabla \cdot \bar{\bar{\tau}})\, dV = \int_\Sigma \vec{r} \wedge (\bar{\bar{\tau}} \cdot \vec{n})\, d\sigma$$

which shows that linear momentum together with a symmetric stress tensor is equivalent to Eq. 3.37. The symmetry of the stress tensor is exactly the condition that makes both statements agree. Note that this is not a derivation of Eq. 3.37 from linear momentum alone: in Notes.pdf (pp. 29-30) the symmetry of $\bar{\bar{\tau}}$ (Eq. 3.19) is itself obtained from angular momentum, so the argument would be circular if read that way.

Step 3: Transfer to a moving control volume (Eq. 3.38)

Apply Eq. 3.8 with the density $\vec{\phi} = \rho\, \vec{r} \wedge \vec{v}$, evaluate the right-hand side of Eq. 3.37 on Σ_c, V_c (coincident with Σ_f, V_f at the instant considered) and split the stress with Eq. 3.28. Since $\vec{r} \wedge \left((-p\bar{\bar{I}}) \cdot \vec{n}\right) = -\vec{r} \wedge (p\vec{n})$:

$$\frac{d}{dt}\int_{V_c} \rho\, \vec{r} \wedge \vec{v}\, dV + \int_{\Sigma_c} \rho\, (\vec{r} \wedge \vec{v}) \left[(\vec{v} - \vec{v}_c) \cdot \vec{n}\right] d\sigma = -\int_{\Sigma_c} \vec{r} \wedge (p\vec{n})\, d\sigma + \int_{\Sigma_c} \vec{r} \wedge (\bar{\bar{\tau}}' \cdot \vec{n})\, d\sigma + \int_{V_c} \rho\, \vec{r} \wedge \vec{f}_m\, dV$$

Eq. 3.38

Because $\vec{x}_0$ is fixed, $\vec{r}$ is not differentiated in the control-volume term; the point $\vec{x}_0$ may be placed anywhere, and the choice is a matter of convenience.

Step 4: Uniform ports and the pivot rule (extension)

Extension, not in the Notes. If a port A_k is small compared with its distance to $\vec{x}_0$, then $\vec{r} \approx \vec{r}_k$ is constant over the port. With uniform ρ_k, $\vec{v}_k$ normal to the port and a fixed volume,

$$\int_{A_k} \rho\, (\vec{r} \wedge \vec{v})(\vec{v} \cdot \vec{n})\, d\sigma \approx \vec{r}_k \wedge \Big[\int_{A_k} \rho \vec{v} (\vec{v} \cdot \vec{n})\, d\sigma\Big] = \dot{m}_k\, \vec{r}_k \wedge \vec{v}_k$$

For steady flow the equation reduces to $\sum_k \dot{m}_k\, \vec{r}_k \wedge \vec{v}_k = \sum \vec{M}_{\vec{x}_0}$, with $\dot{m}_k$ negative at inlets. An unknown reaction force $\vec{R}$ applied at the point $\vec{x}_R$ contributes the moment $(\vec{x}_R - \vec{x}_0) \wedge \vec{R}$, which is zero if $\vec{x}_0 = \vec{x}_R$ (or if $\vec{x}_0$ lies on the line of action of $\vec{R}$). This is the pivot rule.

Step 5: Euler turbomachinery equation (every step)

Extension, not in the Notes. Take a fixed annular control volume around the rotor axis z, with an inlet cylinder $r = r_1$ (normal $-\vec{e}_r$) and an outlet cylinder $r = r_2$ (normal $+\vec{e}_r$), with $\vec{x}_0$ on the axis. The flow is periodic and is treated as steady and axisymmetric on average, with ρ, v_r, v_θ uniform on each cylinder. The end walls are impermeable. The z component of the angular momentum per unit mass follows from $\vec{r} = r\vec{e}_r + z\vec{e}_z$ and $\vec{v} = v_r\vec{e}_r + v_\theta\vec{e}_\theta + v_z\vec{e}_z$:

$$(\vec{r}\wedge\vec{v})_z = r\,v_\theta$$

Convective flux in the z component of Eq. 3.38. On the outlet $\vec{v}\cdot\vec{n} = +v_r$ and on the inlet $\vec{v}\cdot\vec{n} = -v_r$. The mass flow through each cylinder is the same, $\dot m = \int \rho v_r\, d\sigma$, by Eq. 3.11 in steady flow:

$$\int_{\Sigma_0} \rho\,(r v_\theta)(\vec{v}\cdot\vec{n})\,d\sigma = r_2 v_{\theta 2}\,\dot m - r_1 v_{\theta 1}\,\dot m = \dot m\,(r_2 v_{\theta 2} - r_1 v_{\theta 1})$$

Moments on the right-hand side. A pressure force on a cylinder of constant r is $\mp p\,\vec{e}_r\,d\sigma$, and $(r\vec{e}_r + z\vec{e}_z)\wedge\vec{e}_r = z\,\vec{e}_z\wedge\vec{e}_r = z\,\vec{e}_\theta$, which has no z component. On the end walls the force is along $\pm\vec{e}_z$ and $r\vec{e}_r\wedge\vec{e}_z = -r\vec{e}_\theta$, again with no z component. Viscous stresses on the ports and the moment of the body force about the axis are neglected. The only remaining moment is the moment M_z exerted by the blades (solid surfaces inside the volume) on the fluid:

$$M_z = \dot m\,(r_2 v_{\theta 2} - r_1 v_{\theta 1})$$

Here M_z is the torque exerted by the rotor on the fluid. The torque exerted by the fluid on the rotor is $-M_z$ (Newton's third law). Each blade point moves with velocity $\Omega r\,\vec{e}_\theta$, so the power transmitted to the fluid is ΩM_z and the work per unit mass is

$$w_{\mathrm{Euler}} = \frac{\Omega M_z}{\dot m} = u_2 v_{\theta 2} - u_1 v_{\theta 1}, \qquad u = \Omega r$$

Dimensional check: $[\dot m\, r\, v_\theta] = \mathrm{kg\,s^{-1}\,m\,m\,s^{-1}} = \mathrm{N\,m}$ and $[u\,v_\theta] = \mathrm{m^2/s^2} = \mathrm{J/kg}$.

2. Strategic Pivot Selection Strategy

In many exam problems (such as pipe elbows, nozzle mounts, and rotating sprinklers), several complex linear reaction forces $\vec{R}$ act at physical supports, bearings, or hinges. Solving for these linear reaction forces simultaneously with moments can be tedious and prone to algebraic errors.

> THE STRATEGIC PIVOT RULE
>
> Always place the reference reduction point $\vec{x}_0$ **directly at the support hinge, bearing axis, or bolt circle** where unknown linear reaction forces act!
>
> Because the position vector from $\vec{x}_0$ to the reaction point is zero ($\vec{r} = \vec{x} - \vec{x}_0 = \vec{0}$), the moment arm is zero: $\vec{r} \times \vec{R} = \vec{0}$. The unknown linear reaction forces **completely vanish from the moment balance**, allowing direct solution for holding torque or shaft power in a single line!

1D Streams Approximation

For discrete 1D inlets and outlets with mass flow rates $\dot{m}_i$, position vectors $\vec{r}_i = \vec{x}_i - \vec{x}_0$, and mean velocities $\vec{v}_i$, steady angular momentum reduces to:

$$\sum_j \dot{m}_{\text{out},j}(\vec{r}_{\text{out},j} \times \vec{v}_{\text{out},j}) - \sum_k \dot{m}_{\text{in},k}(\vec{r}_{\text{in},k} \times \vec{v}_{\text{in},k}) = \sum \vec{M}_{\vec{x}_0,\text{ext}}$$

CANONICAL WALKTHROUGH 1 · CL3 & CL15 SETUP

Holding Torque on an Offset Pipe Bend / Mixing Elbow

Physical Setup: Consider a horizontal planar pipe bend in the xy-plane. Water enters at inlet 1 located at $\vec{r}_1 = (0, -L_1, 0)$ flowing in $+y$ direction with speed v_1, area A_1, and gauge pressure p_{g1}. It is deflected through 90° and exits at $\vec{r}_2 = (L_2, 0, 0)$ discharging in $+x$ direction with speed v_2, area A_2, at atmospheric pressure ($p_{g2} = 0$). Mass flow rate is $\dot{m} = \rho A_1 v_1$.

The pipe is anchored to a central mounting hub at the origin $\vec{x}_0 = (0, 0, 0)$. Determine the holding torque $\vec{\mathcal{M}}_0 = \mathcal{M}_z \hat{k}$ that the support must exert to prevent the elbow from spinning.

1 Angular Momentum Fluxes

Compute the convective angular momentum cross-products about $\vec{x}_0 = \vec{0}$:

- **Inlet flux:** Position $\vec{r}_1 = -L_1\hat{j}$, velocity $\vec{v}_1 = v_1\hat{j}$.

$$\vec{r}_1 \times \vec{v}_1 = (-L_1\hat{j}) \times (v_1\hat{j}) = \vec{0} \qquad \text{(parallel vectors!)}$$

- **Outlet flux:** Position $\vec{r}_2 = L_2\hat{i}$, velocity $\vec{v}_2 = v_2\hat{i}$.

$$\vec{r}_2 \times \vec{v}_2 = (L_2\hat{i}) \times (v_2\hat{i}) = \vec{0} \qquad \text{(parallel vectors!)}$$

Thus, in this symmetric offset orientation, the convective angular momentum flux is zero! What creates the torque?

2 Moments of External Pressure Forces

Evaluate moments of surface gauge pressure forces:

- **Inlet pressure force:** Pushes inward ($+y$): $\vec{F}_{p1} = p_{g1}A_1\hat{j}$.

$$\vec{r}_1 \times \vec{F}_{p1} = (-L_1\hat{j}) \times (p_{g1}A_1\hat{j}) = \vec{0}$$

- **Outlet pressure force:** Zero because $p_{g2} = 0$ (atmospheric).

General Offset Case: Now suppose the exit is offset by a moment arm d from the x-axis, so $\vec{r}_2 = L_2\hat{i} + d\hat{j}$.

Then $\vec{r}_2 \times \vec{v}_2 = (L_2\hat{i} + d\hat{j}) \times (v_2\hat{i}) = -dv_2\hat{k}$.

The angular momentum balance yields:

$$\dot{m}(-dv_2\hat{k}) = \vec{\mathcal{M}}_{\text{anchor}} \implies \vec{\mathcal{M}}_{\text{anchor}} = -\dot{\mathbf{m}}\mathbf{d}\mathbf{v_2}\hat{\mathbf{k}}$$

The anchor must apply a clockwise holding torque of magnitude $\dot{m}dv_2$ to resist the jet's reaction moment!

CANONICAL WALKTHROUGH 2 · TURBOMACHINERY PRINCIPLE

Euler's Turbomachinery Equation (Rotor Shaft Power)

Problem: A centrifugal compressor or pump rotor rotates at steady angular velocity $\vec{\Omega} = \Omega \hat{e}_z$. Fluid of density ρ enters at inner radius r_1 with mass flow rate $\dot{m}$ and tangential velocity component $v_{\theta 1}$, and exits at outer radius r_2 with tangential velocity component $v_{\theta 2}$. Determine the torque M_z and shaft power $\dot{W}_{\text{shaft}}$ transmitted from the impeller to the fluid.

`Derivation` **Angular Momentum Balance about Axis of Rotation**

Select a fixed cylindrical control volume enclosing the rotating impeller. In cylindrical coordinates (r, θ, z), position vectors from the axis are $\vec{r}_1 = r_1 \hat{e}_r$ and $\vec{r}_2 = r_2 \hat{e}_r$.

The cross product of radius with velocity $\vec{v} = v_r \hat{e}_r + v_\theta \hat{e}_\theta + v_z \hat{e}_z$ isolates the tangential velocity component:

$$\vec{r} \times \vec{v} = (r \hat{e}_r) \times (v_r \hat{e}_r + v_\theta \hat{e}_\theta + v_z \hat{e}_z) = r v_\theta \hat{e}_z - r v_z \hat{e}_\theta$$

Projecting onto the axis of rotation z, the angular momentum per unit mass is simply $r v_\theta$.

For steady flow, the net torque M_z exerted by the rotor blades on the fluid equals the net outflow of angular momentum:

$$\mathbf{M_z = \dot{m}(r_2 v_{\theta 2} - r_1 v_{\theta 1})}$$

Multiplying torque by rotor angular speed Ω gives the shaft mechanical power $\dot{W}_{\text{shaft}}$ delivered to the fluid:

$$\dot{W}_{\text{shaft}} = \Omega M_z = \dot{m}(\Omega r_2 v_{\theta 2} - \Omega r_1 v_{\theta 1})$$

Defining blade peripheral speeds $u_1 = \Omega r_1$ and $u_2 = \Omega r_2$, we obtain the celebrated **Euler Turbomachinery Equation**:

$$\mathbf{w_{\text{Euler}} = \frac{\dot{W}_{\text{shaft}}}{\dot{m}} = u_2 v_{\theta 2} - u_1 v_{\theta 1}}$$

PUMPS VS TURBINES SIGN INTERPRETATION

- **Pumps, Fans, and Compressors ($w_{\text{Euler}} > 0$):** The impeller increases the fluid's whirl ($u_2 v_{\theta 2} > u_1 v_{\theta 1}$). Mechanical shaft work is consumed to raise fluid pressure and stagnation enthalpy.
- **Turbines ($w_{\text{Euler}} < 0$):** The fluid enters with high swirl created by stationary guide vanes/nozzles and leaves with reduced tangential velocity ($u_2 v_{\theta 2} < u_1 v_{\theta 1}$). Fluid delivers mechanical shaft work to drive aircraft generators or compressor spools!